

Conception Phase

DLBDSPBDM01

Build a Data Mart in SQL

Piotr Trybuś

92132989

1. Introduction

This project aims at designing a database for a host-to-guest property booking system. The database is required to store: guest, host, property, review and payment data. This document contains requirements describing the main user groups, their actions and the data.

2. Requirements

2.1 Required Roles

The below key user groups are identified.

- **Guest:** a customer looking to book accommodation.
- **Host:** a user offering properties for short-term rental.
- **Admin:** internal staff, resolving disputes and managing the platform.
- **Payments Provider:** user representing the external payment providers.

2.2 Required Data

The system must store the following categories of data:

- **User and Profile Data:** User identifiers, personal details, role, profile picture, registration timestamp.
- **Host-Related Data:** Host identifiers, contact information, related properties.
- **Property Data:** Property identifiers, names and descriptions, address, amenities and photos, ownership, ratings and reviews.
- **Booking Data:** Guest, host and property references, booking dates, creation date, status, pricing.
- **Payment Data:** Transactions, sender and receiver identifiers, amounts, method and payment status, timestamps for processing and settlement.
- **Communication Data:** Messages between users, sender and receiver identifiers, associated booking, message text and timestamps.

2.3 Required Functions:

Required, core system functions:

- **User Management:** Create, update and retrieve user and role information.
- **Listing Management:** Register and update properties with descriptions.
- **Booking Management:** Create, update, delete bookings, calculate and store pricing information.

- **Payment Processing:** Create payments, update payment status, link transactions to payer and payee accounts.
- **Ratings and Reviews Management:** Store and associate ratings with guests and properties.
- **Messaging:** Store and retrieve messages between hosts and guests.

3. Core Actions For Each Role

3.1 Guest

- Create and maintain a profile.
- Search for properties by filtering through location and amenities.
- Create a booking for a specific property and dates..
- Make payments and receive refunds.
- Send and receive messages with hosts regarding a booking.
- Submit ratings after completing a stay.

3.2 Host

- Create and maintain a profile.
- Create properties, specifying address, amenities, photos and pricing.
- Communicate with guests through messages.
- Receive payments and refunds..

3.4 Payments Provider

- Initiate and update payments.

3.3 Admin

- Review and moderate properties.
- Monitor bookings and payments.
- Manage operations.